

SINGULAR SUPPORT OF ℓ -ADIC SHEAVES

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ABSTRACT. The goal of this seminar is to study the theory of singular support and characteristic cycles of ℓ -adic sheaves as introduced by Beilinson and Saito, respectively. Along the way, we will cover the more classical theory of singular support of sheaves on manifolds introduced by Kashiwara-Schapira.

1. CLASSICAL TOPOLOGICAL SETTING

1.0.1. The subject of microlocal analysis was introduced by Sato in the 1960s. It allows for the measure of not only *where* the singularities of the solution to a partial differential equation occur, but in which codirections.

The theory of microlocal analysis was generalized to sheaves on real manifolds in [KS82], [KS90], which we describe next.

1.0.2. Let e be a field and M a topological manifold. Denote by $\mathrm{Sh}(M)$ the category of sheaves of e -vector spaces on M . We may try to study the topology of M by considering sheaf cohomology:

$$\mathcal{F} \in \mathrm{Sh}(M) \rightsquigarrow R\Gamma(M, \mathcal{F}) \in \mathrm{Vect}.$$

This leads us to study the sheaves themselves. Naturally, we would like to assign invariants to sheaves and ask how these relate to cohomology. Unfortunately, there is not much to say at this level of generality. However, if M is a smooth manifold, we may consider the *singular support* of $\mathcal{F}$, denoted $\mathrm{SS}(\mathcal{F})$, which is a subset of T^*M . This subset is supposed to capture cohomological properties of $\mathcal{F}$ in a precise sense.

1.0.3. The notion of singular support for sheaves on smooth manifolds was introduced by Kashiwara-Schapira in [KS82], [KS90], and its more precise purpose is to measure the codirections where $\mathcal{F}$ is not locally constant.

Before diving into the definition, let us first state its main properties and then provide some examples.

1.0.4. *Properties.* The subset $\mathrm{SS}(\mathcal{F}) \subseteq T^*M$ satisfies:

- (1) It is a closed subset;
- (2) It is stable under the $\mathbb{R}_{>0}$ -action on T^*M ;
- (3) $\mathcal{F}$ is locally constant if and only if $\mathrm{SS}(\mathcal{F})$ is contained in the zero-section 0_M . More generally, if $\mathcal{F}$ is locally constant on an open subset $U \subseteq M$, then $\mathrm{SS}(\mathcal{F})|_{T^*U} \subseteq 0_U$;
- (4) Singular support refines usual support in the sense that $\mathrm{SS}(\mathcal{F}) \cap 0_M = \mathrm{Supp}(\mathcal{F})$;
- (5) (Morse Lemma): Let $\phi : M \rightarrow \mathbb{R}$ be a C^∞ function with $\phi : \mathrm{Supp}(\mathcal{F}) \rightarrow \mathbb{R}$ proper. For $t \in \mathbb{R}$, let $M_t = \phi^{-1}(-\infty, t)$. Let $a < b$ and suppose that $d\phi(x) \notin \mathrm{SS}(\mathcal{F})$ for all $a \leq x < b$. Then the natural map

$$R\Gamma(M_b, \mathcal{F}) \rightarrow R\Gamma(M_a, \mathcal{F})$$

is an isomorphism. In this sense, $\mathrm{SS}(\mathcal{F})$ captures those cotangent directions in which the cohomology of $\mathcal{F}$ changes as we sweep a hypersurface through M .

The above recovers the usual Morse lemma when $\mathcal{F} = \underline{e}_M$.

(6) $\text{SS}(\mathcal{F})$ is coisotropic under the natural symplectic structure on T^*M .¹²

1.0.5. *Definition of singular support.* Let us now give the definition of singular support of sheaves on a smooth manifold M . The basic idea is that $\text{SS}(\mathcal{F})$ is the (closure of the) set of $(x, \xi) \in T^*M$ where the local cohomology around x in the codirection of ξ changes.

Given an open subset B of M , a smooth function $\phi : B \rightarrow \mathbb{R}$ and a point $x_0 \in B$, we define the open subset:

$$U_{x,\phi} = \{x \in B : \phi(x) < \phi(x_0)\} \subseteq M.$$

1.0.6. *Definition.* We have $p = (x_0, \xi_0) \notin \text{SS}(\mathcal{F})$ if and only if there exists an open neighbourhood W of p such that for every smooth function ϕ defined in a neighbourhood of x_0 with $d\phi(x_0) \in W$, the restriction map

$$\text{colim}_V R\Gamma(U_{x_0,\phi} \cup V, \mathcal{F}) \rightarrow R\Gamma(U_{x_0,\phi}, \mathcal{F})$$

is an isomorphism, where V runs over all neighbourhoods of x .

One may imagine sweeping the hypersurfaces $\phi^{-1}(t)$, $t \in \mathbb{R}$ through M , and if $d\phi(x_0) \notin \text{SS}(\mathcal{F})$, then sections $\mathcal{F}$ along the hypersurfaces do not change in a neighbourhood around x_0 .

1.0.7. *Examples.* Let us consider a few examples:

- (1) If $\mathcal{F} = \underline{e}_M$ is the constant sheaf, then by property (3), $\text{SS}(\mathcal{F}) = 0_M$.
- (2) Given $x \in M$, let δ_x be the skyscraper sheaf at x . Then $\text{SS}(\delta_x) = T_x^*M$.
- (3) Let $M = \mathbb{R}$ and $i : [0, \infty) \hookrightarrow \mathbb{R}$ be the natural inclusion. Let $\mathcal{F} = i_!(\underline{e}_{[0,\infty)})$. Then

$$\text{SS}(\mathcal{F}) = \{(x, 0) \mid x \geq 0\} \cup \{(0, \xi) \mid \xi \geq 0\}.$$

It is a good basic exercise to see that $(x, dx) \in \text{SS}(\mathcal{F})$ by considering the function $\phi(x) = x$.

- (4) Let $M = \mathbb{R}$ and $j : \mathbb{R} - \{0\} \hookrightarrow \mathbb{R}$ be the natural inclusion. Let $\mathcal{F} = j_!(\underline{e}_{\mathbb{G}_m})$ be the extension by zero of the constant sheaf on $\mathbb{G}_m$. Then $\text{SS}(\mathcal{F}) = 0_{\mathbb{R}} \cup T_0^*\mathbb{R}$.

2. COMPLEX CONSTRUCTIBLE SETTING

Suppose now that X is a complex manifold. We will mostly be interested in the category of *constructible* sheaves on X .

Concretely, a sheaf $\mathcal{F}$ on X is called constructible if:

- (1) There exists a locally finite stratification $X = \cup_i X_i$ by locally closed submanifolds X_i such that $F|_{X_i}$ is locally constant for all i ;
- (2) The fibers of $\mathcal{F}$ are perfect complexes (bounded complex with finite-dimensional cohomologies).

In the above case, we say that a sheaf $\mathcal{F}$ is constructible against the stratification $\{X_i\}$.

$\text{SS}(\mathcal{F})$ is stable under the $\mathbb{C}^\times$ -action on T^*X .

¹This is surprising: as we will see, there is no a priori reason for the definition of singular support to satisfy symplectic constraints.

²One may draw a loose analogy with Heisenberg's uncertainty principle here. Namely, think of M as spacetime so that points $x \in M$ measure position and cotangent vectors $\xi \in T_x^*M$ measure momentum at x . The coisotropy condition says in particular that $\dim \text{SS}(\mathcal{F}) \geq \dim M$, and so having a small support in M (i.e., a localized position) forces large support in the cotangent directions (i.e., a delocalized momentum).

For example, when $\mathcal{F} = \underline{e}_M$ is the constant sheaf, it is supported entirely on the zero section. On the other hand, when $\mathcal{F} = \delta_x$, the singular support contains the entire cotangent fiber at x .

2.0.1. A key property of constructible sheaves (not immediate from the definition) is that for a constructible sheaf $\mathcal{F}$, there exists a stratification $\{X_i\}$ constructible for $\mathcal{F}$ such that:

$$\mathrm{SS}(\mathcal{F}) \subseteq \bigcup_i T_{X_i}^* X.$$

That is, the singular support of a constructible sheaf is contained in the union of conormal bundles to smooth submanifolds. In particular, $\mathrm{SS}(\mathcal{F})$ is not merely coisotropic but actually Lagrangian.

2.0.2. *Characteristic cycles.* An important refinement of singular support is the *characteristic cycle*. For a constructible sheaf $\mathcal{F}$, its characteristic cycle $\mathrm{CC}(\mathcal{F})$ is a finite formal sum

$$\mathrm{CC}(\mathcal{F}) = \sum_{\alpha} m_{\alpha} \Lambda_{\alpha},$$

where Λ_{α} is an irreducible component of $\mathrm{SS}(\mathcal{F})$. In particular, $\mathrm{Supp}(\mathcal{F}) \subseteq \mathrm{SS}(\mathcal{F})$.

2.0.3. *Index formula.* The following result connects characteristic cycles to cohomology. Suppose $\mathcal{F}$ is a constructible sheaf with compact support. Then:

$$\chi(R\Gamma(X, \mathcal{F})) = (\mathrm{CC}(\mathcal{F}), [0_X])_{T^*X} \in \mathbb{Z}.$$

Here, the right-hand side denotes the intersection of the cycles $\mathrm{CC}(\mathcal{F})$ and $[0_X]$.

3. ÉTALE SETTING

The latter half of this seminar will focus on the theory of singular support and characteristic cycles due to Beilinson and Saito, respectively [Bei16], [Sai17].

3.0.1. Let k be a base field and X a smooth variety over k . Fix a coefficient ring Λ , which we take either to be a finite ring of order invertible in k , or $\Lambda = \overline{\mathbb{Q}}_{\ell}$ for $\ell \neq \mathrm{char}(k)$.

Beilinson defined a theory of singular support for constructible ℓ -adic sheaves with coefficients in Λ .³ Concretely, for $\mathcal{F}$ a constructible sheaf, Beilinson associated a subset

$$\mathrm{SS}(\mathcal{F}) \subseteq T^*X$$

that satisfies many of the properties that singular support does in the complex analytic setting. $\mathrm{SS}(\mathcal{F})$ is:

- (1) closed and conical;
- (2) has dimension equal to $\dim X$;
- (3) $\mathcal{F}$ is lisse if and only if its singular support is contained in the zero section.

The main difference from the classical story is that when $\mathrm{char}(k) > 0$, the singular support no longer needs to be Lagrangian as the following example shows:

Example 3.0.1.1. Let $r : \mathbb{A}_{u,t}^2 \rightarrow \mathbb{A}_{x,t}^2$, $(u, t) \mapsto (u^p + ut^2, t)$, and let $\mathcal{F} = r_*(\mathbb{Q}_{\mathbb{A}_{u,t}^2})$.

r is finite and hence proper. Whatever singular support is, it should satisfy that

$$\mathrm{SS}(r^*(\mathcal{F})) \subseteq r_*(\mathrm{SS}(\mathcal{F})),$$

whenever r is proper, where $r_*(\mathrm{SS}(\mathcal{F}))$ denotes the pull-push of $\mathrm{SS}(\mathcal{F})$ along the correspondence

$$T^*\mathbb{A}_{x,t}^2 \leftarrow T^*\mathbb{A}_{x,t}^2 \times_{\mathbb{A}_{x,t}^2} \mathbb{A}_{u,t}^2 \xrightarrow{dr^*} T^*\mathbb{A}_{u,t}^2.$$

³If Λ is finite, one would usually call these *torsion étale* sheaves rather than ℓ -adic sheaves, but we stick to the latter so that we do not have to distinguish the terminology depending on Λ .

Say $T^*\mathbb{A}_{x,t}^2$ has coordinates (x, ξ, t, τ) . Then

$$dr^*(\xi dx + \tau dt) = \xi t^2 du + (2\xi tu + \tau) dt.$$

It follows that

$$\mathrm{SS}(\mathcal{F}) \subseteq 0_{\mathbb{A}_{x,t}^2} \cup \{t = 0, \tau = 0\}. \quad (3.0.1)$$

Note that r is finite étale of rank p over $t \neq 0$. On the other hand, over $t = 0$ the fibers have cardinality at most 1. So $\mathcal{F}$ is not lisse and hence is not contained in $0_{\mathbb{A}_{x,t}^2}$. Since $\mathrm{SS}(\mathcal{F})$ is closed and conical, it follows that the inclusion (3.0.1) is an equality. But $\{t = 0, \tau = 0\}$ is not Lagrangian.

The above example shows that $\mathrm{SS}(\mathcal{F})$ need not be contained in the union of conormal bundles to subvarieties. It was known for a long time that if a theory of singular support for ℓ -adic sheaves existed, the presence of wild ramification would force such odd behaviour of SS . It was thus a surprise when Beilinson defined a notion of singular support which nonetheless satisfies that $\mathrm{SS}(\mathcal{F}) = \dim X$.

3.0.2. Characteristic cycles and Grothendieck-Ogg-Shafarevich. Suppose k is algebraically closed. Saito [Sai17] defined characteristic cycles for ℓ -adic sheaves. He proved an index formula for ℓ -adic sheaves similar to the one described in 2.0.3:

$$\chi(R\Gamma(X, \mathcal{F})) = (\mathrm{CC}(\mathcal{F}), [0_X])_{T^*X} \in \mathbb{Z}.$$

One may consider this a generalization of the Grothendieck-Ogg-Shafarevich formula to higher dimensions. That is, suppose $j : X = C - \{c_1, \dots, c_n\} \hookrightarrow C$ is the smooth curve obtained by puncturing a smooth projective curve C of genus g at n points, and let $\mathcal{L}$ be a lisse sheaf on X of rank r . Let $\mathcal{F} = j_!(\mathcal{L})$, a constructible sheaf on C . Then:

$$\mathrm{CC}(\mathcal{F}) = -r[0_C] - \sum_{i=1}^n r - \mathrm{Sw}_{c_i}(\mathcal{L}) \cdot [T_{c_i}^*C]$$

where $\mathrm{Sw}_{c_i}(\mathcal{L})$ denotes the Swan conductor of $\mathcal{L}$ at c_i . Combining with the index formula, we obtain the classical Grothendieck-Ogg-Shafarevich formula:

$$\chi(R\Gamma_c(X, \mathcal{L})) = r(2 - 2g - n) - \sum_{i=1}^n \mathrm{Sw}_{c_i}(\mathcal{L}) = r \cdot \chi(H_c^*(X, \overline{\mathbb{Q}}_\ell)) - \sum_{i=1}^n \mathrm{Sw}_{c_i}(\mathcal{L}).$$

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